School of ETMS (Engineering, Technology, Mathematics and Science)
Engineering, Computer Science, and Computer Information Technology
Programming for Engineers
ENGR 2304 3
Lecture
Online

Instructor Contact Information

Roderick Crowder
Email: rcrowder@DallasCollege.edu
Office Hours: MW 1:30-2:00PM; TR 3:30-5:30PM (Virtual)
Office Location: WH184

Course Information

Campus: Online Campus
Course Start and End Date: 08/24/2026 - 12/10/2026
Withdraw Date: 11/12/2026
Certification Date: 09/08/2026

Course Description

Programming principles and techniques for matrix and array operations, equation solving, and numeric simulations applied to engineering problems and visualization of engineering information; platforms include spreadsheets, symbolic algebra packages, engineering analysis software, and laboratory control software.

State-Defined Learning Outcomes

Use matrix and array operations for equation solving. Identify the strengths and weaknesses of the conventional programming languages. Use spreadsheets and their built-in features to solve a variety of engineering problems, applying both quantitative and qualitative methodologies. Describe methods for the design of programs that control equipment or analyze data. Write computer programs to solve engineering problems and perform engineering simulations using common software tools. Graphically present engineering data, results, and conclusions.

Required Course Materials

None

Course Schedule

When	Topic	Notes
Week 1	Course Introduction; Summary of Expressions A02	Ecampus Assignment A01 Due Program P00 Due
Week 2	Summary of Loops, Printf, Scanf A03	Assignment A02 Due Program Assignment P01 Due
Week 3	Summary of Arrays and Functions A04	Assignment A03 Due Program Assignment P02 Due
Week 4	Summary of Pointers A05	Assignment A04 Due Program Assignment P03 Due
Week 5	Strings A06	Assignment A05 Due Program Assignment P04 Due

When	Topic	Notes
Week 6	Summary of Structure, Unions and Enumerations A07	Assignment A06 Due Program Assignment P05 Due
Week 7	Exam 1	Assignment A07 Due Program Assignment P06 Due
Week 8	Complete Assignments / Programming	
Week 9	Summary of C Preprocessor A08	
Week 10	Summary of Structured Files Streaming I/O A09	Assignment A08 Due Program Assignment P07 Due
Week 11	Summary of C Handling A10	Assignment A09 Due Program Assignment P08 Due
Week 12	Exam 2	
Week 13	Complete Assignments / Programming	Assignment A10 Due Program Assignment P09 Due
Week 14	Creating GitHub Repository A11	Assignment A11 Due Program Assignment P10 Due
Week 15	Complete Assignments / Programming	
Week 16	Final Exam	

Reference Calendar for up-to-date with ecampus for current due dates.

Graded Work

Criteria

Type	Weight	Topic	Notes
Assignments	30%	11 Assignments	Weekly Assignments
Programming	50%	10 Programs	
Quizzes	10%	2 Exams (25% each)	
Final Exam	30%	1 Exam	

Breakdown

Grade	Range	Notes
A	90 to 100	
B	80 to 89	
C	70 to 79	
D	60 to 69	
F	below 60	

Course Policies

Unexpected Class Changes

Check ecampus for announcements for current information and changes or class cancellations. Consider installing the Pulse App (Brightspace) on your mobile on instant notification of all announcements

Attendance and Participation

Students are expected to attend all face to face classes and contribute to group work and in-class activities. If you are unable to attend, please inform me. Late Work

Timely and accurate submission of assignments is essential for students to be successful in the course. The assignment due dates for individual assignments will be announced periodically in class. The policy for late submission is as follows:

Time Elapsed After Assigned Due Date	Grade Reduction Penalty
1 day	20%
2 days	40%
3 days	80%
4 days	100% (no credit given)

The date will be based on the upload time to ecampus. An incorrect submission (files incorrectly named or missing elements) will be

considered a non-submission. All assignments are to uploaded into ecampus (Brightspace).

Right to Change and Amend

Instructor retains right to change this syllabus.

Support Contacts

Contact Your Success Coach

Every Dallas College student has a personalized Success Coach who supports them from day one to graduation. Contact your coach for help navigating college and reaching milestones leading to graduation and a career.

Get Free Tutoring

Tutoring is free to all current Dallas College students. You can walk in or schedule an appointment at all Learning Commons campus locations. Live, online tutoring is also available via eCampus.

Explore More Free Student Resources

You have access to many free resources as a Dallas College student, including Counseling and Psychological Services, Child Care Resources, Housing Resources, Emergency Aid Funds, Food Pantries, and more!

Submit the Student Care Form

Not sure which free college resources can help? Submit the Student Care Form! Our Student Care Network will connect you to support for physical and mental health, financial concerns, food, clothing, and more.

Contact Technical Support

Need help with eCampus or another college technology? Our technical support staff can assist you.

Institutional Policies

Please review the **Institutional Policies** page to learn more about accommodations for students with disabilities, class drop and repeat

options, TitleIX (harassment, discrimination, and sexual misconduct), and more.